

Separation of Heart and Lung Sounds by A Deep Network-Based Model

Tung-Sheng Tsai

Ph.D. Program of Electrical and
Communications Engineering,
Feng Chia University
shaiken@fcu.tw

Yi-Jui Liu

Feng Chia University
erliu@fcu.edu.tw

Chun-Chieh Hsu

Feng Chia University
tungsheng999@gmail.com

Ching-Lung Hsieh

Chung Shan Medical University
Hospital
mstts.savior@gmail.com

Abstract—Breath and heart sounds are the two main biological sound signals. Heart sounds are produced by the beating of the heart caused by blood flows in the heart valves and blood vessels, while lung sounds are produced by the flow of air in and out of the airways during the breathing cycle. Both signals are recorded from the chest area. However, these two signals are mostly superimposed on each other, so it is of great research interest to separate these heart sound signals (HSS) and lung sound signals (LSS). A "Looking to Listen" model developed by Google, is a joint audiovisual model for isolating a single speech signal from a mix of sounds such as other speakers and background noise. In this study, we evaluated the feasibility of separating HSS and LSS by the audio part of the "Looking to Listen" model. We collected the pure HSS (PHSS) and pure LSS (PLSS) data from the public website (kaggle dataset, CoronaHack Respiratory Sound Dataset). Then, heart and lung sounds (MHLs) were combined HSS and LSS with different weighting power (3:7, 5:5, and 7:3). A total of 8076 MHLs were generated with 72% of the data in the training set, 10% in the validation set, and 18% in the test set. The cross-correlation coefficient (CC) of pure sound and mixed sound and CC of separated and pure sounds were compared to evaluate the efficacy of extraction HSS or LSS from MHLs. The results showed the mean CC was 0.4, 0.7, and 0.9 for pure HSS and MHLs with 3:7, 5:5, and 7:3, respectively; and the mean CC was 0.79, 0.87, and 0.97 for pure HSS and predicted HSS with 3:7, 5:5, and 7:3, respectively. The mean CC was 0.9, 0.7, and 0.4 for pure LSS and MHLs with 3:7, 5:5, and 7:3, respectively. The mean CC was 0.95, 0.91, and 0.81 for pure LSS and predicted LSS with 3:7, 5:5, and 7:3, respectively. The feasibility of separating HSS and LSS by the audio part of the "Looking to Listen" model was validated.

Keywords—Audio-Visual, Source Separation, Deep Learning, Heart Sound, Lung Sound

I. INTRODUCTION

The heart and lungs are the main organs of the human body, and their sounds reflect the health of the body. The main sound of the heart is called "lub-dub" or "lub-dub-dub" in medical terms. This sound is produced by the opening and closing of the valves that control the flow of blood between each chamber of the heart. Doctors listen to this rhythm to determine normal or abnormal conditions. Abnormal rhythms are a sign of various heart conditions such as atrial fibrillation or valvular disease. It is also important to note that a person's age, gender, and race affect their heart rate. The lungs make several different sounds. These sounds include breath sounds, such as wheezing or popping sounds that occur during inhalation and exhalation, and audible and audible noises

caused by airway obstruction due to mucus or tissue blockage in the airway. Breath sounds indicate that the airways are inflamed from infection or obstruction, reducing oxygen levels in the lungs. Additionally, vocal noises may indicate an upper respiratory disorder such as laryngitis or vocal cord dysfunction syndrome (VCD). However, these two signals are mostly superimposed on each other, so it is of great research interest to separate these heart sound signals (HSS) and lung sound signals (LSS). HSS and LSS separation was first studied in 1986, and since then, in several studies, algorithms to effectively separate HSS and LSS have been developed. In traditional research, basic high-pass frequency filters, adaptive filtering, wavelet denoising algorithms, time-frequency filters, modulation filters, blind source separation (BSS) techniques, and independent component analysis (ICA) were used. Cervera et al. [1] and Macgregor et al. [2] developed an algorithm to separate snoring noise from HSS. As a type of artificial intelligence (AI), deep learning (DL) can be applied to the analysis of complex data of complex computer vision, natural language understanding, and powerful machine learning. In this study, we evaluated the feasibility of separating HSS and LSS by the audio part of the "Looking to Listen" model. We collected the pure HSS (PHSS) and pure LSS (PLSS) data from the public website (kaggle dataset, CoronaHack Respiratory Sound Dataset).

II. RELATED WORKS

Related research on heart and lung sounds in speech separation and audio-visual signal processing was reviewed through a literature review. Heart sound signals have been extensively researched over the past few decades. Automated heart sound segmentation and classification technology have the potential to classify diseases in various applications.

Cervera et al. [1] and Macgregor et al. [2] separated the snoring sounds from the HSS for apnea-related sleep. Wang and Chen used DL-based audio-only methods that addressed speech denoising and speech separation tasks. Gabbay et al. [3] trained an enhancement model on videos where other speech samples of the target speaker were used as background noise. Gabbay et al. [4] used a video-to-sound synthesis method to filter noisy audio. Afouras et al. [5] used a network to predict the magnitude and phase of a denoised speech spectrogram. Ephrat and Google Research [6] separated audio and visual data for the development of the audio-only model.

We used the "Looking to Listen" audio-visual model to separate HSS and LSS to find available sounds MHLs datasets from the public website to train the model.

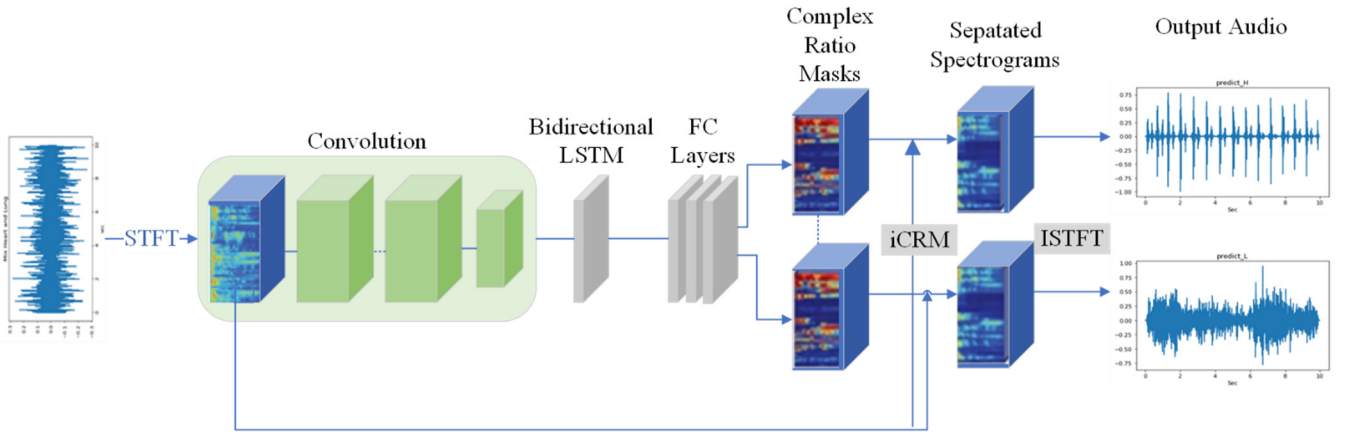


Fig. 1. Model's neural network-based architecture: The MHLS takes as input the soundtrack, containing a mixture of HS and LS noise. And learn a visual feature using a dilated convolutional NN. Audio stream first computes the STFT of the input signal to obtain a spectrogram, and then learns an audio representation using a similar dilated convolutional NN. A joint, audio-visual representation is then created by concatenating the learned visual and audio features, and is subsequently further processed using a bidirectional LSTM and three fully connected layers.

III. DATASET

We searched for available datasets and collected the pure HSS (PHSS) and pure LSS (PLSS) data from public websites (Kaggle dataset [7], CoronaHack Respiratory Sound Dataset [8]). The existing data sets have different lengths, and only the lengths greater than 10 seconds are kept. All the data were pre-processed as follows.

A. Same Audio Length

All audio longer than 13 s was split into another 10-s audio in 3-s intervals (13-s audio will be split into one from 0 to 10 ss, and another from 3 to 13 ss.)

B. Same Frequency

The frequency was unified to 4000.

C. Mixed Dataset

To make the model better segmented in any situation, the HSS and LSS were mixed (MHLS) in different ratios of 3:7, 5:5, and 7:3. A total of 8076 MHLSs were generated with 72% of the data in the training set, 10% in the validation set, and 18% in the test set.

IV. EXPERIMENT

The host graphics card was used to handle graphics processing tasks to improve graphics processing performance. For DL, an efficient graphics card trained models to speed up the interaction between the model and data. Large-scale data and multi-layer neural network training was greatly accelerated by using host graphics cards. Compared with CPUs, GPUs have a higher degree of parallelism and speed up loops running on different layers on large data sets. The host graphics card accelerates visualization or neural network design1, and training significantly speeds up learning to obtain excellent and efficient performance. In the host

computer, CPU i7-12700, GPU RTX3070 8G, and Memory 64GB were used.

V. METHOD

Noisy audio was used as input, and outputs complex spectrogram masks, HS, and LS (Fig. 1).

A. Steps of Model

The noisy input spectrogram was multiplied by the mask to obtain the isolated signal while suppressing all other interfering signals. At inference time, the separation model was applied to arbitrarily segment MHLS in a long unit. Masking with cRM, denoised waveforms were obtained by performing inverse STFT (ISTFT) on the complex multiplication of the predicted cRM and noisy spectrogram. The output of the model was a multiplicative spectrogram mask, which described the time-frequency relationships of clean speech to background interference. We built an Anaconda isolated development environment on Windows and used Tensorflow (tf-2.3) for DL to train the model.

VI. RESULT AND DISCUSSION

A total of 8076 MHLSs were generated with 72% of the data in the training set, 10% in the validation set, and 18% in the test set. The cross-correlation coefficient (CC) of pure and mixed sounds and CC of separated and pure sounds were compared for the efficacy of extraction HSS or LSS from MHLS to evaluate the model.

Cross-correlation between $\{X_i\}$ and $\{X_j\}$ was defined by the ratio of covariance to root-mean variance,

$$\rho_{i,j} = \frac{r_{i,j}}{\sqrt{\sigma_i^2 \sigma_j^2}} \quad (1)$$

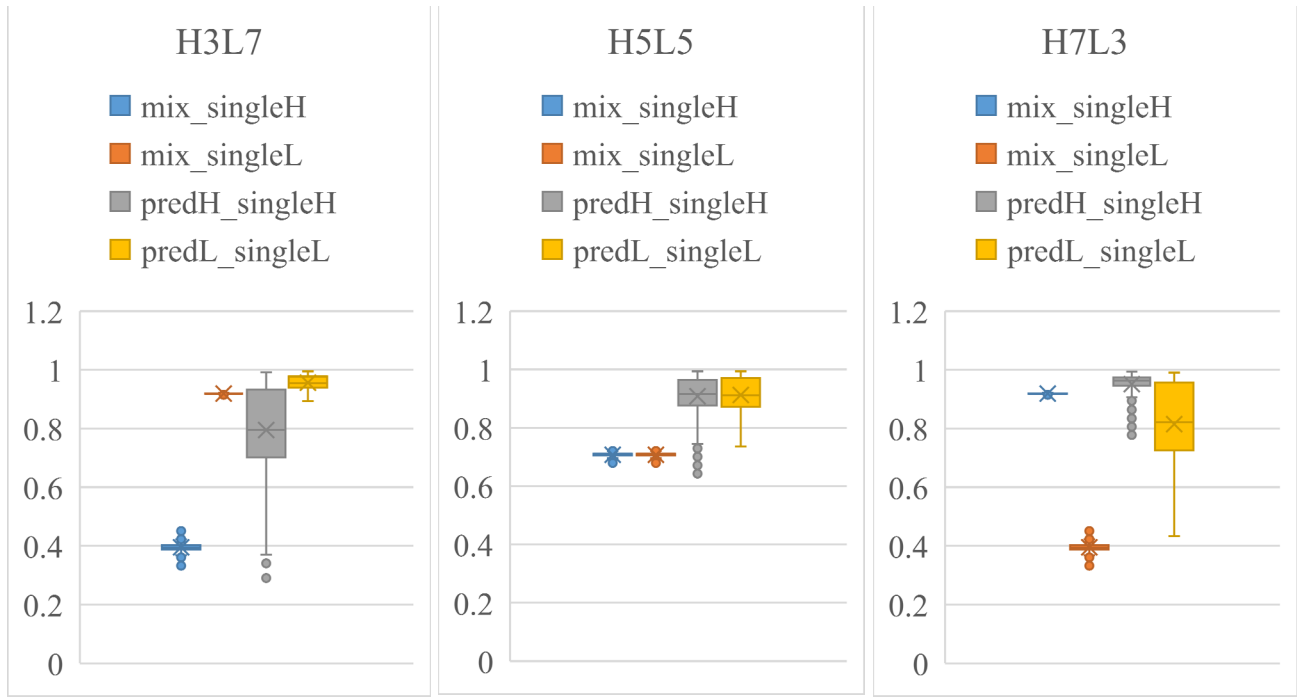


Fig. 2. **Means CC of Total:** The mean CC was 0.4, 0.7, and 0.9 for pure HSS and MHLS with 3:7, 5:5, and 7:3, respectively; and the mean CC was 0.79, 0.87, and 0.97 for pure HSS and predicted HSS with 3:7, 5:5, and 7:3, respectively. The mean CC was 0.9, 0.7, and 0.4 for pure LSS and MHLS with 3:7, 5:5, and 7:3, respectively; and the mean CC was 0.95, 0.91, and 0.81 for pure LSS and predicted LSS with 3:7, 5:5, and 7:3, respectively.

Sample covariance was calculated as

$$\hat{\gamma}_{i,j} = \frac{1}{N} \sum_{t=1}^N [(x_i^t - x_i)(x_j^t - x_j)] \quad (2)$$

Sample cross-correlation was defined by the ratio as follows.

$$\hat{\rho}_{i,j} = \frac{\sum_{t=1}^N [(x_i^t - x_i)(x_j^t - x_j)]}{\sqrt{\sum_{t=1}^N (x_i^t - x_i)^2 \sum_{t=1}^N (x_j^t - x_j)^2}} \quad (3)$$

The mean CC results are shown in Tables I and II. The mean CC was 0.4, 0.7, and 0.9 for pure HSS and MHLS with 3:7, 5:5, and 7:3, respectively; and the mean CC was 0.79, 0.87, and 0.97 for pure HSS and predicted HSS with 3:7, 5:5, and 7:3, respectively. The mean CC was 0.9, 0.7, and 0.4 for pure LSS and MHLS with 3:7, 5:5, and 7:3, respectively; and the mean CC was 0.95, 0.91, and 0.81 for pure LSS and predicted LSS with 3:7, 5:5, and 7:3, respectively.

TABLE I. MEAN CC OF SOUNDS

MHLS Ratio	HSS/MHLS	HSS/Pred HSS	LSS/MHLS	LSS/Pred LSS
3:7	0.4	0.79	0.9	0.95
5:5	0.7	0.87	0.7	0.91
7:3	0.9	0.97	0.4	0.81

VII. CONCLUSION

Separation of heart and lung sounds by a deep network-based model in a single channel was performed by the proposed model in this study. To train the model, we collected the available heart and lung sound datasets from the website and merged all the data before processing. The longer heart and lung sounds were extracted and mixed as the cross-correlation coefficient for verification. The results

demonstrated the feasibility of separating HSS and LSS by the audio part of the "Looking to Listen" model.

REFERENCES

- [1] Morales Cervera S, Nikoli D, Barney A, et al. A model of breathing abnormalities in sleep for development of classification and diagnosis techniques. Paper presented at: 3rd IEEE International Symposium on Applied Sciences in Biomedical and Communication Technologies (ISABEL 2010); 2010 Nov 7–10; Rome. p. 1–2.
- [2] Macgregor CA, Karimi D, Azarbarzin A, et al. Statistical analysis of tracheal breath sounds during wakefulness for screening obstructive sleep apnea. Paper presented at: 35th Annual International Conference of the IEEE EMBS; 2013 Jul 3–7; Osaka, Japan.
- [3] Aviv Gabbay, Asaph Shamir, and Shmuel Peleg. 2017. Visual Speech Enhancement using Noise-Invariant Training. arXiv preprint arXiv:1711.08789 (2017).
- [4] Aviv Gabbay, Ariel Ephrat, Tavi Halperin, and Shmuel Peleg. 2018. Seeing Through Noise: Speaker Separation and Enhancement using Visually-derived Speech. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (2018).
- [5] T. Afouras, J. S. Chung, and A. Zisserman. 2018. The Conversation: Deep Audio-Visual Speech Enhancement. In arXiv:1804.04121.
- [6] Ariel Ephrat, Inbar Mosseri, Oran Lang, Tali Dekel, Kevin Wilson, Avinatan Hassidm, William T. Freeman, Michael Rubinstein, ACM Trans. Graph., Vol. 37, No. 4, Article 112. Publication date: August 2018.
- [7] Kaggle Dataset, Respiratory Disease Classification LSTM/GRU/CNN, <https://www.kaggle.com/code/pvnbalarammurthy/r-espiratory-disease-classification-lstm-gru-cnn/input>
- [8] Kaggle Dataset, CoronaHack Respiratory Sound Dataset, <https://www.kaggle.com/datasets/praveengovi/coronahack-respiratory-sound-dataset>